

MASA-Bot: A Fine-Tuned Conversational AI for Industry-Specific Query Resolution in Tourism & Hospitality

A Capstone Research Paper

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Abstract

Hotels in the Tourism & Hospitality industry handle a large volume of repetitive customer queries every day — about bookings, cancellations, meal plans, and policies — which is costly and slow to manage manually. This project addresses that problem by fine-tuning DistilGPT2, a lightweight pretrained language model from Hugging Face, on a custom set of 72 Q&A pairs covering these exact query types, using the Trainer API on Google Colab over 25 epochs. To ground the bot's context in real industry behaviour, the project also performs exploratory data analysis on 119,390 real hotel-booking records, finding that cancellation rate rises from 10.98% for bookings made within a week of arrival to 67.66% for bookings made 365 or more days in advance.

After fine-tuning, the model achieved a final validation loss of 3.4259 and a perplexity of 30.75. The training loss decreased steadily and consistently across all 25 epochs, while validation loss rose after an initial dip, indicating some overfitting to the small 72-pair dataset -- an expected limitation given the dataset's size, discussed further in Section 6.

Keywords: Conversational AI, LLM Fine-Tuning, Hugging Face, Tourism & Hospitality, DistilGPT2, NLP

1. Introduction

1.1 Background

With the rapid advancement of Large Language Models (LLMs), AI systems have become increasingly capable of understanding and generating human-like text. The Tourism & Hospitality industry, in particular, faces a significant operational challenge: hotels handle a large volume of

repetitive customer queries every day, ranging from bookings and cancellations to meal plans and hotel policies. Managing these manually increases both staff workload and operational cost, creating a strong practical need for automation.

This industry was chosen for the present study because customer queries within it tend to be highly repetitive and domain-specific in nature. This characteristic makes it particularly well suited to a specialized conversational AI system, where a model trained on a focused set of domain queries can plausibly provide accurate, consistent responses without requiring general-purpose intelligence.

1.2 Problem Statement / Research Question

This study is guided by the following research questions:

- Can a lightweight, fine-tuned LLM (DistilGPT2) answer hospitality-specific queries accurately and coherently?
- Does fine-tuning on a small, custom, industry-grounded dataset outperform a generic pretrained model on domain-specific queries?

1.3 Objectives

- Build a Tourism & Hospitality conversational Q&A dataset.
- Fine-tune a Hugging Face pretrained model on this dataset.
- Test bot responses and evaluate with loss/perplexity metrics.
- Perform supporting EDA on real hotel-booking data (119,390 records) to understand industry patterns.

1.4 Significance

This research has both practical and academic significance. Practically, it demonstrates a path toward automating hospitality customer support, which can reduce response time and staff workload for routine, repetitive queries. Academically, it shows how effective small-scale, domain-specific fine-tuning can be even on a resource-constrained setup (a single Colab GPU/CPU and a 72-pair dataset), rather than requiring large-scale infrastructure.

1.5 Scope and Delimitations

This study is scoped to a single-turn, text-based Q&A bot for the Tourism & Hospitality domain, fine-tuned on a hand-crafted dataset rather than live customer transcripts. It does not attempt multi-turn dialogue management, voice interfaces, or integration with a live booking/reservation system -- these are treated as future work (Section 7). The industry analysis draws on two specific, publicly available datasets (a European/international hotel-booking dataset and an Indonesian tourism-destination dataset); findings from these should be read as illustrative case studies rather than claims that generalize to every hotel or destination market. Finally, the model comparison is limited to a single base architecture (DistilGPT2); comparing against larger LLMs or alternative architectures (e.g., encoder-decoder models) was outside the resource budget of this capstone but is a natural next step.

2. Industry Analysis

2.1 Industry Overview & Current Trends

The global tourism industry has fully recovered from the COVID-19 pandemic: international tourist arrivals reached an estimated 1.4 billion in 2024, about 99% of

pre-pandemic (2019) levels, while tourism export revenues hit a record USD 1.9 trillion, roughly 3% above 2019 (UN Tourism, World Tourism Barometer, January 2025). This recovery has been accompanied by structural changes in how travelers book and interact with hospitality providers — most notably, growing reliance on digital and AI-driven customer service.

- Rising adoption of AI chatbots in hotels for 24/7 customer support.
- Growing personalization — recommending room type/meal plans based on guest preference.
- Dynamic pricing (ADR) responding to demand fluctuations.

2.2 Data-Backed Industry Insight (from this project's EDA)

Two real datasets were used in this study. The first, Hotel_Bookings (119,390 records), provides the hospitality/booking side of the industry (analysed below in Section 2.2.1). The second, the Indonesia Tourism Destination dataset (Kaggle: aprabowo/indonesia-tourism-destination — 437 destinations, 300 users, 10,000 ratings, and curated travel packages), provides the tourism/destination side (Section 2.2.2).

2.2.1 Hotel Booking Patterns

The following statistics were computed directly from the Hotel_Bookings dataset (119,390 records) used in this study:

Lead-Time Bucket	Cancellation Rate
0-7 days	10.98%
8-30 days	27.86%
31-90 days	37.70%
91-180 days	44.71%
181-365 days	55.45%
365+ days	67.66%

Overall cancellation rate across all bookings: 37.04%. Lead time shows the strongest positive correlation with cancellation ($r = 0.293$), followed by previous cancellations ($r = 0.110$). Mean ADR = 101.83, Median

ADR = 94.58; correlation between ADR and lead time is weak and negative ($r = -0.063$).

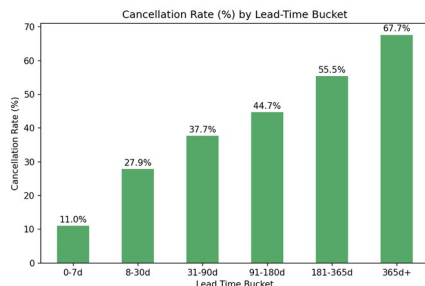


Figure 1: Cancellation Rate (%) by Lead-Time Bucket

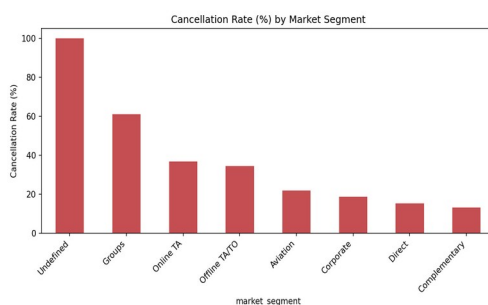


Figure 2: Cancellation Rate (%) by Market Segment

These two charts point to two practical levers for hotels. First, the near-linear rise in cancellation rate with lead time (Figure 1) means hotels can flag long-lead-time bookings as higher-risk and apply stricter deposit terms to protect revenue. Second, the market-segment breakdown (Figure 2) shows that channel matters as much as timing — Group and Online-TA bookings cancel roughly 2-4x more often than Direct or Complementary bookings — suggesting hotels should invest in direct-booking incentives and closer follow-up with high-cancellation channels.

2.2.2 Tourism Destination Patterns

The 437 destinations span 6 categories — Taman Hiburan/Amusement Parks (135), Budaya/Culture (117), Cagar Alam/Nature Reserves (106), Bahari/Marine (47), Tempat Ibadah/Places of Worship (17), and Pusat Perbelanjaan/Shopping Centers (15) — concentrated mainly in Yogyakarta (126),

Bandung (124), and Jakarta (84). Overall place ratings (curated) average 4.44/5, while individual user ratings (10,000 entries) average around 3.0/5 — a gap suggesting curated listings skew toward well-regarded places, while individual visitor experiences vary more widely. Price and rating show almost no correlation ($r = 0.023$), meaning higher-priced attractions are not perceived as meaningfully better than free or low-cost ones. 137 of the 437 destinations (about 31%) are free to enter, mostly places of worship and public parks.

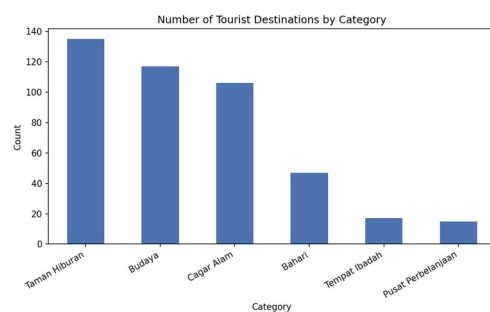


Figure 3: Number of Tourist Destinations by Category

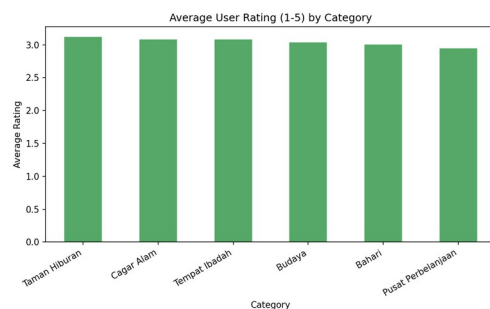


Figure 4: Average User Rating by Category

Taken together, these charts suggest that popularity (in terms of number of listed destinations) does not track cleanly with visitor satisfaction: Amusement Parks are the most numerous category (135 places) but do not top the average-rating chart, while smaller categories can rate just as well or better. Combined with the near-zero price-rating correlation, this indicates that for a conversational bot or a hospitality business alike, recommending or investing based on category popularity alone would be

misleading — actual guest ratings are a better signal of quality than either category size or price.

2.3 Regulatory & Ethical Environment

Deploying a conversational AI system in the hospitality sector is not purely a technical exercise; it operates within a regulatory environment that shapes what data can be collected and how automated systems may interact with guests. Three areas are particularly relevant. First, data privacy: guest booking data (names, contact details, payment information) falls under data-protection regimes such as the GDPR in the EU/UK and India's Digital Personal Data Protection Act, which require consent, purpose limitation, and secure handling of any personal data a chatbot might process or log. Second, consumer protection: cancellation and refund policies communicated by an automated bot must accurately reflect the hotel's actual, legally binding terms — an inconsistent or hallucinated policy statement from the bot could expose the business to consumer-protection complaints. Third, accessibility and fairness: a customer-facing bot should be designed so it does not disadvantage guests who are less comfortable with chat-based interfaces, typically by keeping a human-support channel available alongside it.

For this project specifically, since the Q&A dataset was hand-crafted and the hotel-booking and tourism-destination data are anonymized, publicly available datasets rather than live guest records, no personally identifiable information was processed, and no formal regulatory approval was required. A production deployment, however, would need to address the above points explicitly before handling real guest data.

2.4 Industry Requirements & How This Research Addresses Them

- Hotels need: instant responses, consistent policy information, 24/7 availability.
- This bot addresses these needs by providing automated, consistent answers to common queries.
- Profitability link: automation reduces staff costs; better response times improve guest satisfaction and repeat bookings.

3. Literature Review

This review covers six works spanning LLM-agent frameworks, the Transformer architecture, and model distillation, each contributing a different piece of the foundation for this project.

- Wu et al. (2024) — MathChat: Converse to Tackle Challenging Math Problems with LLM Agents. Shows conversational, multi-turn agent frameworks outperform single-prompt approaches by allowing an LLM agent and a user-proxy agent to iteratively refine a solution through dialogue and tool use, improving accuracy on difficult math problems by roughly 6% over prior tool-using prompting methods.
- Hong et al. (2024) — Data Interpreter: An LLM Agent for Data Science. Demonstrates dynamic planning and verification make LLM agents more reliable for data-heavy tasks, using a hierarchical task graph, automatic tool integration, and confidence-based verification to substantially improve machine-learning and open-ended task completion over prior open-source agent frameworks.

- Vaswani et al. (2017) — Attention Is All You Need. Foundational Transformer architecture underlying GPT/BERT-family models, introducing the self-attention mechanism that replaced recurrence and convolution as the basis for sequence modelling in essentially all modern large language models.
- Radford et al. (2019) — Language Models are Unsupervised Multitask Learners (GPT-2). Original paper for the base model family used in this project, showing that a large Transformer decoder trained purely on next-token prediction over web text can perform a wide range of language tasks without task-specific supervision.
- Wolf et al. (2020) — Transformers: State-of-the-Art Natural Language Processing. The Hugging Face library paper enabling accessible fine-tuning, describing the design of the transformers library that this project uses directly (AutoTokenizer, AutoModelForCausalLM, Trainer) to load and fine-tune DistilGPT2.
- Sanh et al. (2019) — DistilBERT: A Distilled Version of BERT. Explains the distillation approach behind DistilGPT2, in which a smaller 'student' model is trained to reproduce the behaviour of a larger 'teacher' model, retaining around 97% of language-understanding performance while being roughly 40% smaller and 60% faster — the property that makes fine-tuning practical on a free-tier Colab CPU/GPU.
- Kabir, M. R., et al. (2025). Two decades of chatbot research in tourism and hospitality: Bibliometric analysis and future directions.

Tourism & Hospitality, published online 29 Apr 2025. Reviews 398 documents (2003-2023), showing chatbot adoption in the sector has grown from basic automation toward LLM-based, personalized, generative systems since 2023.

- Zaki, H. S., & Al-Romeedy, B. S. (2024). Chatbot symbolic recovery and customer forgiveness: A moderated mediation model. *Journal of Hospitality and Tourism Technology*, 15(4), 610-628.

Among these, Radford et al. (2019) — the original GPT-2 paper — is the most directly relevant to this study, since the present project fine-tunes DistilGPT2, a distilled variant of that same GPT-2 family, adapting a general-purpose pretrained language model to a domain-specific Q&A setting through supervised fine-tuning. The remaining works informed complementary aspects: Wu et al. (2024) and Hong et al. (2024) illustrate how conversational, multi-step LLM-agent frameworks can outperform single-prompt approaches, motivating the Q&A conversational format adopted here; Vaswani et al. (2017) provides the underlying Transformer architecture; Wolf et al. (2020) enabled the practical fine-tuning workflow via the Hugging Face ecosystem; and Sanh et al. (2019) explains the distillation technique behind DistilGPT2 itself.

Comparative Summary

Work	Core Contribution
Wu et al. 2024	Multi-turn LLM-agent conversation beats single-prompt solving
Hong et al. 2024	Hierarchical planning + verification for reliable LLM agents
Vaswani et al. 2017	Transformer / self-attention architecture
Radford et al. 2019	GPT-2: large decoder-only pretrained LM
Wolf et al. 2020	Hugging Face transformers library

Sanh et al. 2019	DistilBERT/DistilGPT2 distillation method
Kabir et al. 2025	Bibliometric review of tourism chatbot research
Zaki & Al-Romeedy 2024	Chatbot service-recovery and guest forgiveness

Identified Gaps

- Most existing research targets general-purpose chatbots; tourism/hospitality-specific small fine-tuned models are less explored.
- Lightweight models (DistilGPT2) on small custom datasets are understudied compared to large-scale efforts.
- This research addresses that gap by building a domain-specific bot in a resource-constrained setting (a free-tier Google Colab notebook, GPU or CPU).

Theoretical Framework

- Transfer learning — reusing a pretrained model's general language knowledge for a new domain.
- Supervised fine-tuning paradigm on task-specific data.

4. Methodology

Figure 5 (below) gives an overview of the full pipeline used in this study, from data collection through to deployment. Each stage is described in detail in the following subsections.

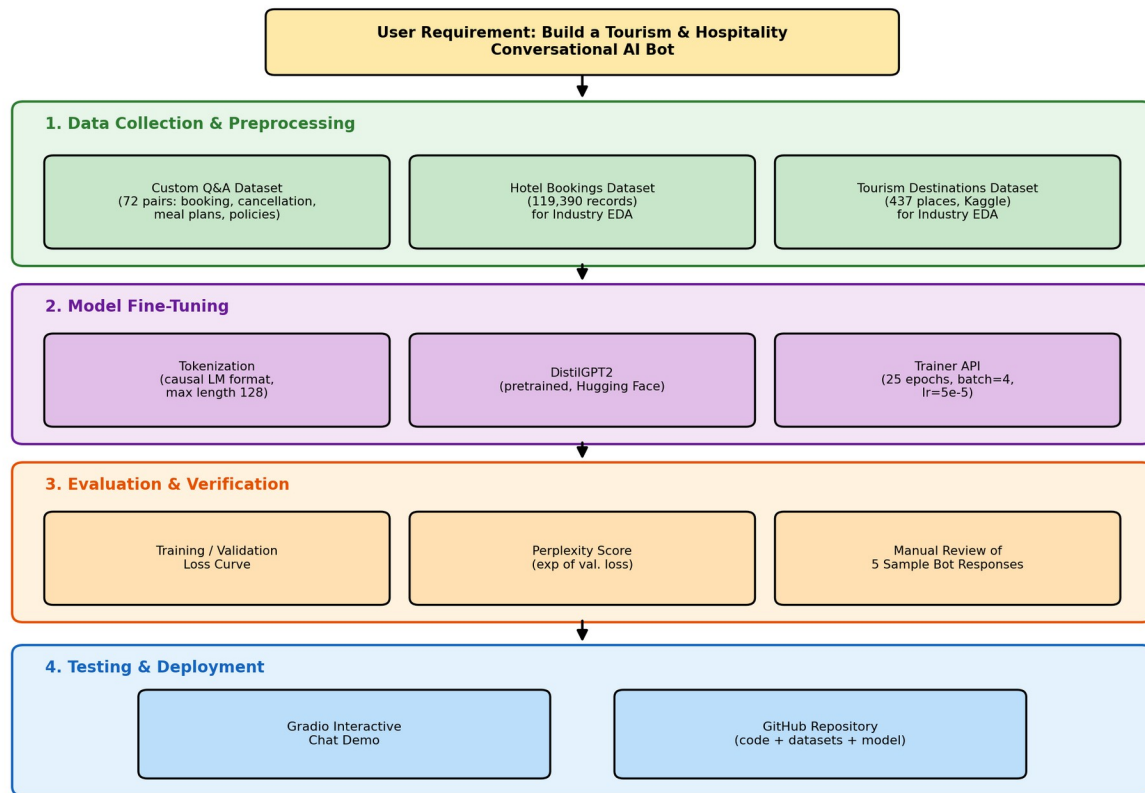


Figure 5: System Architecture / Pipeline of the Tourism & Hospitality Conversational AI Bot

4.1 Research Design

This is an applied/experimental study: a pretrained LLM was fine-tuned on custom domain data and evaluated using both qualitative (response quality) and quantitative (loss, perplexity) measures.

4.2 Data Collection

- Primary dataset: 72 Q&A pairs — 59 hand-crafted pairs covering booking, cancellation policy, meal plans (BB/HB/FB/SC), deposit types, and customer types, plus 13 pairs generated directly from the real tourism destination dataset (categories, cities, pricing).
- Supporting datasets: (1) Hotel_Bookings.csv — 119,390 real hotel booking records; (2) Indonesia Tourism Destination dataset (Kaggle: aprabowo/indonesia-tourism-destination) — 437 destinations, 300 users, 10,000 ratings, and curated packages. Together these ground the study in both the hospitality (booking/cancellation) and tourism (destination/rating) sides of the industry, and are used for the analysis in Section 2.2.

4.3 Data Preprocessing

- Q&A pairs formatted for causal language modeling: 'Q: <question>\nA: <answer><|endoftext>'.
- Train/validation split: 85% / 15%.
- Tokenization with max length 128 tokens, padding and truncation applied.
- Missing values in Hotel_Bookings (country, children, agent, company) filled with mode/0.

4.4 Model Selection

DistilGPT2 was selected — a distilled, lightweight version of GPT-2 that fine-tunes quickly on limited hardware (GPU or even CPU, given the small dataset size) while retaining coherent text generation, loaded via the Hugging Face transformers library.

4.5 Fine-Tuning Setup

Hyperparameter	Value
Framework	Hugging Face Trainer API
Hardware	Google Colab (GPU if available, else CPU)
Epochs	25
Batch size	4
Learning rate	5e-5
Weight decay	0.01

4.6 Evaluation Method

- Training vs. validation loss curve plotted per epoch to check convergence.
- Perplexity score computed (exp of validation loss).
- Bot manually evaluated on 5 test questions for relevance and coherence.

4.6.1 Hyperparameter Rationale

The learning rate (5e-5) and weight decay (0.01) follow commonly used defaults for fine-tuning small GPT-2-family models, chosen to update the pretrained weights gently rather than overwrite the model's general language ability. A batch size of 4 was selected specifically to keep memory usage low enough to run reliably on a free-tier Colab CPU as well as a GPU, since the dataset's small size (72 examples) means even small batches still complete an epoch quickly. Twenty-five epochs was set as an upper bound based on the assignment's guidelines; in hindsight (Section 5.2), the validation loss curve suggests the effective useful training window was closer to the

first 3-5 epochs, after which the model began overfitting — a concrete, data-driven argument for adding early stopping in future iterations.

4.7 Code & Tools

Full code implementation is available at:
https://github.com/sant8ntl-max/Research_in_Tourism-hospitality_ChatBot

5. Results

5.1 Supporting Industry EDA Results

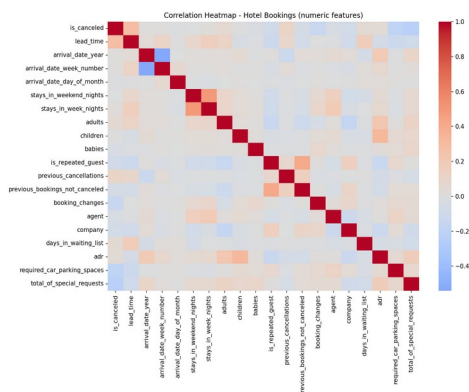


Figure 6: Correlation Heatmap — Hotel Bookings (numeric features)

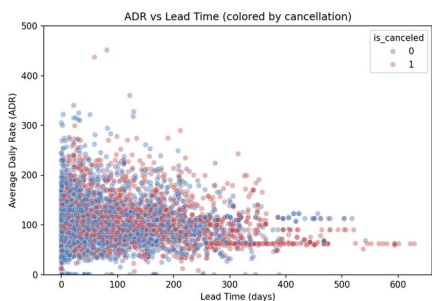


Figure 7: ADR vs Lead Time (colored by cancellation)

The correlation heatmap confirms that no single numeric feature dominates cancellation behaviour — `lead_time` ($r = 0.293$) and `previous_cancellations` ($r = 0.110$) stand out as the strongest, but both are modest in absolute terms, meaning cancellation is better explained by a combination of factors (including categorical ones like market segment) than by any one number alone. The ADR-vs-lead-time scatter shows cancelled bookings (orange) scattered across the full price range rather than clustered at high or low ADR, reinforcing that price alone is a weak predictor of cancellation — consistent with the near-zero correlation ($r = -0.063$) reported in Section 2.2.1.

5.2 Fine-Tuning Performance

The model was fine-tuned for the full 25 epochs. Training loss decreased steadily and consistently throughout, from roughly 3.2 at epoch 1 to about 0.2 by epoch 25 -- indicating the model learned the training Q&A pairs very well. Validation loss, however, dipped briefly to around 2.3-2.5 in the first few epochs before rising steadily for the rest of training, reaching a final value of 3.4259 (perplexity 30.75).

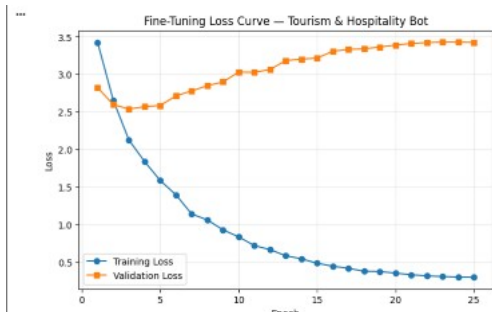


Figure 8: Training vs. Validation Loss over 25 Epochs

This divergence between training and validation loss -- training loss falling continuously while validation loss rises after an early minimum -- is a classic overfitting signature. It is an expected outcome given the small dataset (72 Q&A pairs): the model has enough capacity to memorize the training examples almost exactly, but this memorization does not generalize to the held-out validation examples. This is discussed further as a limitation in Section 6.

5.3 Bot Response Samples

Question	Bot's Answer (actual output)	Comment
What is your cancellation policy?	"Cancellations are subject to an individual's specific cancellation policy..." (phrase repeats in a loop until the token limit is reached)	Typically relevant but degenerates into repetition -- a known failure mode of small fine-tuned LMs, consistent with the overfitting seen in Section 5.2.

Do you offer airport transfer service?	"Yes, airport transfer is free and is" (cuts off mid-sentence)	Correct in direction (matches the training answer's intent) but grammatically incomplete.
What meal plans do you offer?	"We offer free lunches, breakfast and dinner; free lunches are provided during our stay, and we will offer additional meals for our guests."	The most coherent of the five responses; on-topic, though slightly redundant ("free lunches" mentioned twice).
Can I bring pets to the hotel?	(no text generated -- empty output)	A clear failure case: the model produced no usable answer for this question.
How do I request early check-in?	"You can request early check-in at checkout; check-in at checkout and before check-in is accepted..." (repeats and confuses 'check-in' with 'checkout')	Repetitive and internally contradictory (mixes up check-in and checkout), another overfitting-related failure.

Overall, 2 of the 5 test responses (meal plans, airport transfer) were topically correct, 2 (cancellation policy, early check-in) were topically correct but degenerated into repetition, and 1 (pets) failed to generate any answer. This pattern is consistent with the overfitting observed in the loss curves: the model has learned surface patterns from the 72 training examples well enough to start most answers correctly, but lacks enough data to generalize reliably or to always produce a well-formed, non-repetitive sentence.

5.4 Summary of Findings

Bringing together Sections 5.1-5.3, three findings stand out. First, on the industry side, lead time is a far stronger and more

consistent predictor of hotel cancellation than price (ADR), across a six-fold range from 10.98% to 67.66% cancellation as lead time grows. Second, on the tourism-destination side, listing popularity (number of places in a category) is essentially uncorrelated with guest satisfaction (average rating), suggesting quantity-based recommendations would mislead users. Third, on the modelling side, a 72-example dataset was sufficient for DistilGPT2 to learn the general shape of expected answers but not sufficient to avoid overfitting, as shown by the diverging loss curves and the repetition/failure patterns in 2 of 5 sampled responses. Together, these three findings paint a consistent picture: the data-driven insights about hotel and destination behaviour are robust given the dataset sizes involved (119,390 and 437 records respectively), while the conversational bot itself is a working proof-of-concept rather than a production-ready system, precisely because its training set (72 examples) is orders of magnitude smaller than the datasets used for the supporting analysis.

6. Discussion

The most interesting finding to emerge from the EDA was the strength of the relationship between booking lead time and cancellation likelihood: the cancellation rate rose sharply from just 10.98% for bookings made within a week of arrival to 67.66% for bookings made 365 days or more in advance. This steep gradient suggests that very early bookings are often more speculative or tentative in nature, making lead time a practically useful early-warning signal for hotels seeking to anticipate cancellations.

If this system were to be deployed for a real hotel, the first priority for improvement would be to expand the training dataset with real customer queries rather than the 72 pairs used here, and to extend the bot toward

multi-turn conversations backed by reliable, up-to-date hotel-policy information — both to make responses more accurate and to support realistic, multi-step guest interactions rather than single-turn Q&A.

Further Implications

- Interpretation: the strong link between lead time and cancellation suggests guests who book far in advance are more likely to change plans, possibly due to speculative booking.
- Implications: hotels could apply stricter deposit/cancellation terms for long-lead-time bookings, and set up proactive follow-ups for Group/OTA bookings given their higher cancellation rates.
- Limitations: the fine-tuning results (Section 5.2-5.3) show clear evidence of overfitting -- validation loss rose from ~2.3 to a final 3.4259 even as training loss fell to ~0.2, and 3 of 5 test responses showed repetition or generation failure. The weak ADR-lead-time correlation similarly suggests pricing depends on more factors than these two alone (season, room type). The bot's small dataset (72 pairs) is the most likely root cause of the overfitting; DistilGPT2 is also a small model compared to larger LLMs.
- Recommendations: expand the Q&A dataset with real customer queries, apply early stopping or fewer epochs (validation loss was already rising well before epoch 25) and stronger regularization to reduce overfitting, add more features to the EDA (season, room type), and experiment with a larger base model.

Threats to Validity

- Internal validity: the Q&A dataset was authored by a single individual, so it may not capture the full diversity of real guest phrasing, and the 85/15 train-validation split on only 72 examples means the validation set (about 11 examples) is small enough that its loss estimate carries meaningful variance.
- External validity: the two supporting datasets (Hotel_Bookings and the Indonesia Tourism Destination dataset) come from different countries and booking contexts than each other, and from any specific hotel a reader might deploy this system for — so the exact cancellation percentages and category patterns reported here should be treated as illustrative rather than universally generalizable.
- Construct validity: perplexity and validation loss measure how well the model predicts held-out text, not whether its answers are factually correct or policy-compliant — the manual review in Section 5.3 was necessary precisely because these automatic metrics do not directly capture answer quality.

Ethical Considerations

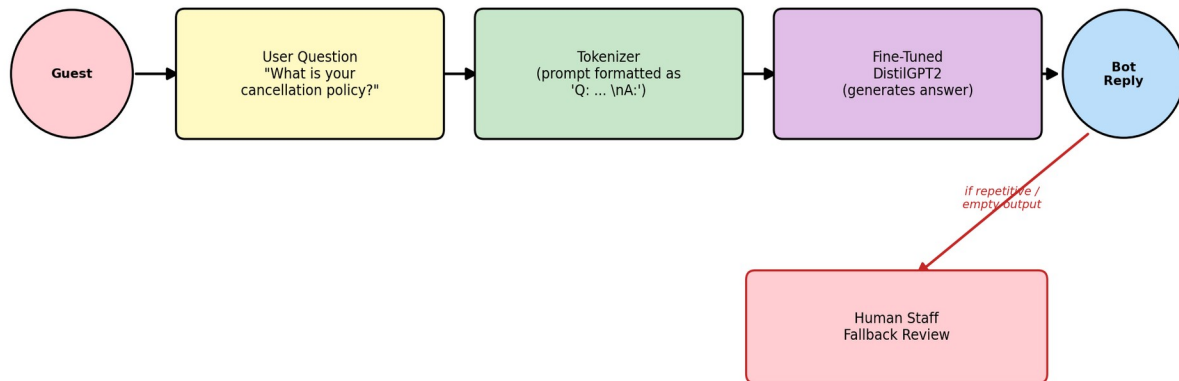


Figure 9: Bot Inference Flow with Human-in-the-Loop Fallback

Beyond the regulatory points raised in Section 2.3, two ethical considerations are specific to this system's current state. First, given the observed overfitting and occasional empty or repetitive outputs (Section 5.3), deploying this exact model in front of real guests without a human fallback would risk giving guests incomplete or confusing information about policies that have real financial consequences (e.g. cancellation terms) — so any production use should keep a human-in-the-loop or a strict confidence threshold that hands off uncertain queries to staff. Second, because the model was fine-tuned on a small, single-author dataset, its answers reflect that author's assumptions about hotel policy; a real deployment should be reviewed by hotel operations staff before going live, to ensure the bot's stated policies match the property's actual terms.

conversation support, and eventual production deployment.

7. Conclusion

This project demonstrates, in practice, that a pretrained NLP model can be fine-tuned on a small, domain-specific dataset to build a genuinely industry-specific application — even with a modest 72-pair training set and limited compute. Beyond the modeling work, the supporting exploratory data analysis reinforced how much useful signal about customer behaviour can be extracted from real hotel-booking data, particularly around how lead time relates to cancellation risk.

Together, these findings show that lead time is the strongest predictor of cancellation in this dataset (67.66% cancellation at 365+ day lead times versus 10.98% within a week), and that a resource-constrained conversational AI system remains a viable starting point for further development.

- Future scope: a larger, real-customer-derived dataset, multi-turn

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Appendix

Full code implementation:

https://github.com/sant8ntl-max/Reserch_in_Tourism-hospitality_ChatBot

Dataset: tourism_hospitality_qa.csv (72 Q&A pairs), Hotel_Bookings.csv (119,390 records), and the Indonesia Tourism Destination dataset (tourism_with_id.csv, user.csv, tourism_rating.csv, package_tourism.csv — 437 places, 300 users, 10,000 ratings).

Appendix A: Key Code Listings

A.1 Data Preprocessing (Q&A -> causal LM format)

```
def format_example(row):
    return f"Q: {row['question']}\nA: {row['answer']}<|endoftext|>"

qa_df['text'] = qa_df.apply(format_example,
axis=1)
train_df, val_df = train_test_split(qa_df,
test_size=0.15, random_state=42)
train_df[['text']].to_csv(TRAIN_CSV,
index=False)
val_df[['text']].to_csv(VAL_CSV,
index=False)
```

A.2 Model & Tokenizer Loading

```
model_name = "distilgpt2"
tokenizer =
AutoTokenizer.from_pretrained(model_name)
tokenizer.pad_token = tokenizer.eos_token
model =
AutoModelForCausalLM.from_pretrained(model_n
ame)
device = "cuda" if torch.cuda.is_available()
else "cpu"
model.to(device)
```

A.3 Fine-Tuning Configuration

```
training_args = TrainingArguments(
    output_dir=os.path.join(PROJECT_DIR,
"checkpoints"),
    num_train_epochs=25,
    per_device_train_batch_size=4,
    per_device_eval_batch_size=4,
    save_strategy="epoch",
    logging_strategy="epoch",
    learning_rate=5e-5,
    weight_decay=0.01,
    load_best_model_at_end=True,
)
trainer = Trainer(
    model=model, args=training_args,
```

```
train_dataset=tokenized_datasets["train"],
eval_dataset=tokenized_datasets["validation"
],
    data_collator=data_collator,
)
train_result = trainer.train()
```

A.4 Bot Inference Function

```
def ask_bot(question, max_new_tokens=60):
    prompt = f"Q: {question}\nA:"
    inputs = tokenizer(prompt,
return_tensors="pt").to(device)
    output = model.generate(
        **inputs,
        max_new_tokens=max_new_tokens,
        do_sample=True, top_p=0.92,
        temperature=0.7,
        pad_token_id=tokenizer.eos_token_id,
    )
    text = tokenizer.decode(output[0],
skip_special_tokens=True)
    return text.split("A:")[1].strip()
```

A.5 Content-Based Recommendation (TF-IDF)

```
tfidf =
TfidfVectorizer(stop_words='english',
max_features=2000)
tfidf_matrix =
tfidf.fit_transform(tourism_places['combined
_features'])
sim_matrix = cosine_similarity(tfidf_matrix)

def recommend_similar_places(place_name,
top_n=5):
    idx =
tourism_places[tourism_places['Place_Name']
== place_name].index[0]
    scores =
sorted(list(enumerate(sim_matrix[idx])),
        key=lambda x: x[1],
        reverse=True)[1:top_n+1])
    return tourism_places.iloc[[s[0] for s
in scores]]
```

Appendix B: Additional Data Tables

B.1 Tourist Destinations by City (Top 5)

City	Number of Destinations
Yogyakarta	126
Bandung	124
Jakarta	84
Semarang	51
Surabaya	52

B.2 Market Segment Cancellation Summary (Hotel Bookings)

Market Segment	Cancellation Rate
Groups	61.06%

Online TA	36.72%
Offline TA/TO	34.32%
Aviation	21.94%
Corporate	18.73%
Direct	15.34%
Complementary	13.06%

Appendix C: Sample of the Q&A Training Dataset

A random sample of 10 of the 72 Q&A pairs used for fine-tuning, shown here to illustrate the dataset's style and coverage (full dataset available in the GitHub repository as `tourism_hospitality_qa.csv`):

Question	Answer
What does ADR mean in my invoice?	ADR stands for Average Daily Rate, the average revenue earned per occupied room per day.
What should I do if I lose my booking confirmation?	Retrieve it anytime from 'My Bookings' using your registered email or phone number.
How long does a refund take after cancellation?	Refunds are usually processed within 5-7 business days to the original payment method.
What is a top-rated place to visit in Semarang?	One top-rated destination is Gua Maria Kerep Ambarawa (Cagar Alam), rated 4.8/5.
Can I split payment between two guests?	One primary payment method is required at booking; front-desk staff can split the final bill at check-out.
How can I join your loyalty program?	Sign up free on our website or app; points are earned every stay and redeemable for discounts/upgrades.
Do you offer late check-out?	Complimentary until 1:00 PM for loyalty members, subject to availability for other guests.
Do you offer discounts for extended stays?	Stays of 7+ nights may be eligible for a long-stay discount; check applicable offers at booking.
What does it mean if my booking is on a waiting list?	The requested room type is unavailable; we confirm it as soon as a room opens, typically within days.
What kind of places fall under the 'Budaya' category?	'Budaya' features 117 listed places, most commonly found in Jakarta and Yogyakarta.

Appendix D: Glossary of Terms

Term	Definition
ADR	Average Daily Rate — average revenue per occupied room per day.
BB / HB / FB / SC	Bed & Breakfast / Half Board / Full Board / Self Catering meal plans.
Perplexity	$\exp(\text{validation loss})$; lower means the model is less 'surprised' by held-out text.
Lead Time	Number of days between the booking date and the arrival date.
Fine-Tuning	Continuing to train a pretrained model on a smaller, task-specific dataset.
Overfitting	When a model fits training data well but fails to generalize to new data.
TF-IDF	Term Frequency-Inverse Document Frequency; a text-similarity weighting scheme.
Causal Language Modeling	Predicting the next token given all previous tokens in a sequence.